FREE GROUPS AND FREE ABELIAN GROUPS AS FILTERED COLIMITS

Abstract. In univalent foundations with set quotients, the free group $F(X)$ and the free abelian group $A(X)$ on an arbitrary set X are constructed as filtered colimits of the free structures on the finite stages $f : [n] \to X$. On a finite set the free structures have decidable normal forms (reduced words, integer vectors), and the colimit transfers what these normal forms say to an arbitrary X. We obtain the universal properties, the injectivity of $\eta : X \to F(X)$, and, for a locally small set E, that $F(E)$ is locally small and that η has small fibres. The construction is written once, for any family of free structures on finite stages satisfying a short list of properties; groups and abelian groups are two instances. Everything is checked in Unit.

1. Introduction

We work in univalent type theory with function extensionality, propositional truncation $\|-\|$ and set quotients by proposition-valued equivalence relations. Set quotients are effective: if $[a] = [b]$ then $a \sim b$.

On a set with decidable equality the free group is the group of reduced words, and equality of elements is decidable. On an arbitrary set X reducedness still makes sense, but a word can no longer be reduced, since we cannot decide whether a letter is followed by its inverse. The quotient of all words by the equivalence relation generated by cancellation gives the universal property, but the universal property alone does not give an explicit criterion for equality in $F(X)$; the colimit gives one directly.

We use instead that a set is a filtered colimit of finite sets. A *stage* is a finite set $[n]$ with a map $f : [n] \to X$, and $F(X)$ is the colimit of the free groups $F[n]$. Two words over stages become equal in $F(X)$ exactly when some further stage identifies them, and at a stage equality is decidable. This answers questions such as: what is the fibre of η at $e_0 - e_1 + e_2$ in the free abelian group, when equality on E is not decidable? Equality in $F(E)$ is a statement about a function between finite sets, equalities in E and an equality of normal forms, so it is small when E is locally small; and $\eta x = z$ forces x to be one of the finitely many generators occurring in a representative of z.

Results. For any set X:

(A) $F(X)$ is a group with $\eta : X \to F(X)$, and every map $X \to G$ to a group extends uniquely to a homomorphism; likewise $A(X)$ for abelian groups (Theorems 5.3, 5.4).
(B) η is injective (Theorem 6.1).
(C) If E is locally small, then $F(E)$ and $A(E)$ are locally small and η has small fibres (Theorem 7.3).

2. Finite stages

The finite types we use are given by codes: 0, 1, and $a + b$ for codes a, b, with $\mathrm{El}(0) = \emptyset$, $\mathrm{El}(1) = \{*\}$, $\mathrm{El}(a+b) = \mathrm{El}(a) \sqcup \mathrm{El}(b)$. Sums of stages are then sums of codes, and no arithmetic on $[n]$ is needed. Every $\mathrm{El}(c)$ has a decidable equality, by induction on c. Two further facts are proved by induction on the code:

- $\mathrm{El}(c)$ is inhabited or empty;

- for a decidable predicate p on $\mathrm{El}(c)$, either $p(i)$ for some i, or $p(i)$ fails for every i.

3. The free group on a discrete type

Let D be a type with decidable equality. A letter is a or a^{-1} for $a \in D$; a word is a list of letters; it is *reduced* if no letter is followed by its inverse. Reducedness is decidable.

The letter x acts on words by

$$\mathrm{act}_x(\mathrm{nil}) = x, \qquad \mathrm{act}_x(y\,w) = \begin{cases} w & \text{if } y = x^{-1}, \\ x\,y\,w & \text{otherwise,} \end{cases}$$

and a word $u = x_1 \cdots x_n$ acts by $\mathrm{ev}_u = \mathrm{act}_{x_1} \circ \cdots \circ \mathrm{act}_{x_n}$. Both preserve reduced words. Following van der Waerden [1], the group laws come from the action.

Lemma 3.1. *For reduced s, t, w, any letter x and any words u, r:*

(C) $\mathrm{act}_x(\mathrm{act}_{x^{-1}} w) = w$;
(K) $\mathrm{ev}_{\mathrm{act}_x r}(t) = \mathrm{act}_x(\mathrm{ev}_r t)$;
(A) $\mathrm{ev}_{\mathrm{ev}_u s}(t) = \mathrm{ev}_u(\mathrm{ev}_s t)$;
(N) $\mathrm{ev}_w(\mathrm{nil}) = w$;
(I) $\mathrm{ev}_{u^{-1}}(\mathrm{ev}_u s) = s = \mathrm{ev}_u(\mathrm{ev}_{u^{-1}} s)$, *where u^{-1} is the reversed word of inverses.*

Proof. (C) If $w = \mathrm{nil}$ or w does not start with x, then $\mathrm{act}_{x^{-1}} w = x^{-1}w$ and act_x cancels it. If $w = x\,w'$, then $\mathrm{act}_{x^{-1}} w = w'$, and w' does not start with x^{-1} since w is reduced, so $\mathrm{act}_x w' = x\,w'$. (K) If $r = y\,r'$ with $y = x^{-1}$, the left side is $\mathrm{ev}_{r'} t$ and the right side is $\mathrm{act}_x \mathrm{act}_{x^{-1}} \mathrm{ev}_{r'} t$, equal by (C). Otherwise $\mathrm{act}_x r = x\,r$ and both sides are $\mathrm{act}_x(\mathrm{ev}_r t)$. (A) By induction on u, using (K). (N) By induction on w: $\mathrm{act}_x w' = x\,w'$ when $x\,w'$ is reduced. (I) By induction on u, using (C) and $\mathrm{ev}_{uv} = \mathrm{ev}_u \circ \mathrm{ev}_v$. □

Proposition 3.2. *The reduced words form a group $F(D)$ with $u \cdot v = \mathrm{ev}_u(v)$, unit* nil, *inverse* $\mathrm{ev}_{u^{-1}}(\mathrm{nil})$. *Its generators $a \mapsto a$ are distinct.*

Proof. Associativity is (A), the left unit holds by definition, the right unit is (N), and the inverse laws follow from (I) with (A) and (N). If the one-letter words a and b are equal, a occurs in b, so $a = b$. □

Proposition 3.3 (universal property)**.** *Let G be a group and $f : D \to G$. Put $\Phi(x_1 \cdots x_n) = f(x_1) \cdots f(x_n)$, with $f(a^{-1}) = f(a)^{-1}$. Then Φ is a homomorphism on $F(D)$ extending f, and every homomorphism ψ extending f equals Φ.*

Proof. $\Phi(\mathrm{act}_x r) = f(x)\Phi(r)$ because $f(x)f(x^{-1}) = 1$; by induction $\Phi(\mathrm{ev}_u s) = \Phi(u)\Phi(s)$. For uniqueness, a reduced word $x\,w$ is the product of the one-letter word x and w, and $\psi(a^{-1}) = \psi(a)^{-1}$; induct on the word. □

Proposition 3.4 (functoriality and support)**.** *A map $s : D \to D'$ of discrete types gives the homomorphism $s_* : F(D) \to F(D')$, $s_*(u) =$ the reduced form of s applied to u. Then $(t \circ s)_* = t_* \circ s_*$, $\mathrm{id}_* = \mathrm{id}$, and s_* depends only on the values of s. If a generator b occurs in $s_* u$, then $b = s(a)$ for a generator a occurring in u. Moreover $\Phi_f \circ s_* = \Phi_{f \circ s}$.*

Proof. As (K): $\mathrm{ev}_{s(\mathrm{act}_x r)}(t) = \mathrm{act}_{s(x)}(\mathrm{ev}_{sr} t)$, since $s(x^{-1}) = s(x)^{-1}$. Hence $\mathrm{ev}_{s(\mathrm{ev}_u v)}(t) = \mathrm{ev}_{su}(\mathrm{ev}_{sv} t)$, which gives both the homomorphism property (with (A)) and functoriality. For the support: the letters of $\mathrm{act}_x w$ are among x and the letters of w, so the letters of a reduced form are among those of the word. □

4. The free abelian group on a finite type

For a code c let $\mathbb{Z}^c$ be the set of functions $\mathrm{El}(c) \to \mathbb{Z}$, an abelian group under pointwise addition, with the generators δ_j ($\delta_j(i) = 1$ if $i = j$ and 0 otherwise). We write $[P]\,x$ for x if the decidable statement P holds and 0 otherwise.

Finite sums. In an abelian group G, the sum of $g : \mathrm{El}(c) \to G$ is defined by induction on the code: $\sum_0 g = 0$, $\sum_1 g = g(*)$, $\sum_{a+b} g = \sum_a g \circ \mathrm{inl} + \sum_b g \circ \mathrm{inr}$.

Lemma 4.1. (1) $\sum_i (g(i) + h(i)) = \sum_i g(i) + \sum_i h(i)$ *and* $\sum_i 0 = 0$.
(2) *(Fubini)* $\sum_{i \in \mathrm{El}(c)} \sum_{j \in \mathrm{El}(d)} g(i,j) = \sum_j \sum_i g(i,j)$.
(3) *(δ-rule)* $\sum_i [i = w]\, h(i) = h(w)$ *for* $w \in \mathrm{El}(c)$.

Proof. All three by induction on the code. For (1) on $a + b$ one uses $(x + y) + (z + w) = (x + z) + (y + w)$. For (2) on $a + b$, the induction hypotheses and (1). For (3) on $a + b$ with $w = \mathrm{inl}\, w'$: the terms on the b side are 0, since $\mathrm{inr}\, j \neq \mathrm{inl}\, w'$, and the a side is $h(\mathrm{inl}\, w')$ by induction. □

Multiples. For $g \in G$ put $n \cdot g = g + \cdots + g$ (n times) and $(-n-1) \cdot g = -((n+1) \cdot g)$.

Lemma 4.2. (1) $(x + y) \cdot g = x \cdot g + y \cdot g$, *hence* $(\sum_i n_i) \cdot g = \sum_i n_i \cdot g$.
(2) *A homomorphism* $\chi : \mathbb{Z} \to G$ *is* $n \mapsto n \cdot \chi(1)$.

Proof. (1) First $(x + 1) \cdot g = g + x \cdot g$ and $(x - 1) \cdot g = -g + x \cdot g$, by cases on x, using $-(a+b) = -a - b$. Addition on $\mathbb{Z}$ adds y to x by $|y|$ successors or predecessors, and induction along this gives (1). (2) $\chi(0) = 0$, $\chi(x + 1) = \chi(x) + \chi(1)$ and $\chi(x - 1) = \chi(x) - \chi(1)$; induct on n and on $-n - 1$. □

The groups $\mathbb{Z}^c$. For $m : \mathrm{El}(a) \to \mathrm{El}(b)$ put

$$m_* u\,(z) = \sum_{m(i)=z} u(i) = \sum_i [m(i) = z]\, u(i).$$

Proposition 4.3. m_* *is a homomorphism;* $(n \circ m)_* = n_* \circ m_*$, $\mathrm{id}_* = \mathrm{id}$, m_* *depends only on the values of* m, *and* $m_* \delta_i = \delta_{m(i)}$. *The generators are distinct. If* $\delta_j = m_* u$, *then* $j = m(i)$ *for some* i *(support).* $\mathbb{Z}^0$ *is trivial.*

Proof. m_* is additive by Lemma 4.1(1). For functoriality,

$$\begin{aligned} n_*(m_* u)(w) &= \sum_z [n(z) = w] \sum_i [m(i) = z]\, u(i) = \sum_i \sum_z [m(i) = z]\, [n(z) = w]\, u(i) \\ &= \sum_i [n(m(i)) = w]\, u(i), \end{aligned}$$

by Fubini and the δ-rule; $\mathrm{id}_* u = u$ and $m_* \delta_i = \delta_{m(i)}$ are instances of the δ-rule. If $\delta_i = \delta_j$ then $\delta_i(j) = 1$, so $i = j$. If $\delta_j = m_* u$, then $1 = \sum_i [m(i) = j]\, u(i)$; decide whether some i has $m(i) = j$: if none, every term is 0 and so is the sum. □

Proposition 4.4 (universal property)**.** *Let* G *be an abelian group and* $f : \mathrm{El}(c) \to G$. *Then* $\Phi_f(u) = \sum_i u(i) \cdot f(i)$ *is a homomorphism* $\mathbb{Z}^c \to G$ *with* $\Phi_f(\delta_i) = f(i)$; *every homomorphism* ψ *with* $\psi(\delta_i) = f(i)$ *equals* Φ_f; *and* $\Phi_f \circ m_* = \Phi_{f \circ m}$.

Proof. Φ_f is additive by Lemmas 4.1(1) and 4.2(1); $\Phi_f(\delta_i) = \sum_k [k = i]\, f(k) = f(i)$. Naturality:

$$\Phi_f(m_* u) = \sum_z \sum_i \big([m(i) = z]\, u(i)\big) \cdot f(z) = \sum_i \sum_z [m(i) = z]\, u(i) \cdot f(z) = \sum_i u(i) \cdot f(m(i)),$$

by Lemma 4.2(1), Fubini and the δ-rule. Uniqueness, by induction on the code. On 0, $\mathbb{Z}^0$ is trivial and $\psi(0) = 0$. On 1, $\mathbb{Z}^1 = \mathbb{Z}$ and $\psi(n) = n \cdot \psi(1) = n \cdot f(*)$ by Lemma 4.2(2). On $a + b$, every u is $\mathrm{inl}_* u_a + \mathrm{inr}_* u_b$ with $u_a = u \circ \mathrm{inl}$, $u_b = u \circ \mathrm{inr}$ (by the δ-rule at each point), and $\psi \circ \mathrm{inl}_*$ is a homomorphism $\mathbb{Z}^a \to G$ sending δ_i to $f(\mathrm{inl}\, i)$; by induction it is $\Phi_{f \circ \mathrm{inl}}$, and likewise on b. So $\psi(u) = \Phi_{f\circ\mathrm{inl}}(u_a) + \Phi_{f\circ\mathrm{inr}}(u_b) = \Phi_f(u)$. □

5. The colimit

The construction uses only the following data on the finite stages: for each code c a group $T(c)$ with generators $\mathrm{El}(c) \to T(c)$; for each map m a homomorphism m_*, functorial, depending only on the values of m, sending generators to generators; distinct generators; the support property; $T(c)$ trivial when $\mathrm{El}(c)$ is empty; and the universal property of $T(c)$ for a class of target groups, natural in c ($\Phi_f \circ m_* = \Phi_{f\circ m}$). Sections 3 and 4 give two instances: $T(c) = F(\mathrm{El}c)$ for all groups and $T(c) = \mathbb{Z}^c$ for abelian groups.

Fix a type X. A *stage* is $s = (c, f)$ with $f : \mathrm{El}(c) \to X$; a map of stages $(c, f) \to (d, g)$ is $m : \mathrm{El}(c) \to \mathrm{El}(d)$ with $g \circ m = f$. Stages have sums $(c, f) + (d, g) = (c + d, [f, g])$.

Lemma 5.1 (coequaliser)**.** *Let $g : \mathrm{El}(n) \to X$ and let $(p_i, q_i)_{i\in\mathrm{El}(d)}$ be pairs in $\mathrm{El}(n)$ with $g(p_i) = g(q_i)$. There is $r : \mathrm{El}(n) \to \mathrm{El}(n)$ with $g \circ r = g$ and $r(p_i) = r(q_i)$ for all i.*

Proof. Merge the pairs one at a time, starting from the identity. To merge (a, b) after r_0, put $r = \sigma \circ r_0$ with $\sigma(z) = r_0(a)$ if $z = r_0(b)$, and $\sigma(z) = z$ otherwise. Then $g \circ r = g$ because $g(r_0 a) = g(a) = g(b) = g(r_0 b)$; $r(a) = r_0(a) = r(b)$; and pairs already merged by r_0 stay merged, as r factors through r_0. □

An *element* is a stage s with $u \in T(s)$. Two elements are *linked*, $\mathrm{Link}((s, u), (s', v))$, if there are a stage t and maps $m : s \to t$, $n : s' \to t$ with $m_* u = n_* v$.

Lemma 5.2. *The relation $\|\mathrm{Link}\|$ is an equivalence relation.*

Proof. Reflexivity and symmetry are clear. Given links (t_1, m_1, n_1) from x to y and (t_2, m_2, n_2) from y to z, go to $t_1 + t_2$ and apply Lemma 5.1 to the pairs $(\mathrm{inl}\, n_1(j), \mathrm{inr}\, m_2(j))$, j in the stage of y; these have the same image in X. With the resulting r, $(r \,\mathrm{inl}\, m_1)_* u = (r \,\mathrm{inl}\, n_1)_* v = (r \,\mathrm{inr}\, m_2)_* v = (r \,\mathrm{inr}\, n_2)_* w$. □

Let $F(X)$ be the quotient of the elements by $\|\mathrm{Link}\|$, and $\eta(x) = [(1, x), \mathrm{generator}]$.

Theorem 5.3. *$F(X)$ is a group with*

$$[(s, u)] \cdot [(s', v)] = [(s + s',\ \mathrm{inl}_* u \cdot \mathrm{inr}_* v)], \quad 1 = [(0, 1)], \quad [(s, u)]^{-1} = [(s, u^{-1})].$$

In the abelian instance it is commutative.

Proof. Links are compatible with the operations: from links (t, m, n) and (t', m', n') one gets the link $(t+t', m+m', n+n')$ between the products, since $(m+m')_*(\mathrm{inl}_* u \cdot \mathrm{inr}_* v) = \mathrm{inl}_* m_* u \cdot \mathrm{inr}_* m'_* v$. So the operations descend to the quotient, and the laws, which are equations in a set, may be checked on representatives. There they hold up to a single map of stages: the associativity map $(s + s') + s'' \to s + (s' + s'')$, the maps $0 + s \to s$ and $s + 0 \to s$, and the folding map $s + s \to s$ together with $0 \to s$ for the inverse laws. In the abelian instance the swap $s + s' \to s' + s$ links $x \cdot y$ to $y \cdot x$, since T is commutative at each stage. □

Theorem 5.4 (universal property)**.** *Let G be a target group and $f : X \to G$. There is a homomorphism $\bar{f} : F(X) \to G$ with $\bar{f} \circ \eta = f$, and every homomorphism ψ with $\psi \circ \eta = f$ equals $\bar{f}$.*

Proof. Put $\bar{f}[(s,u)] = \Phi_{f\circ s}(u)$. Along a map of stages $m : s \to t$, naturality gives $\Phi_{f\circ t}(m_* u) = \Phi_{f\circ t\circ m}(u) = \Phi_{f\circ s}(u)$, so $\bar{f}$ is constant on links and descends. It is a homomorphism because each Φ is, and $\bar{f}(\eta x) = f(x)$. For uniqueness, restrict ψ to a stage s: $u \mapsto \psi[(s,u)]$ is a homomorphism on $T(s)$ (the folding map $s + s \to s$ links $(s,u)\cdot(s,v)$ to (s,uv)), sending the generator i to $\psi(\eta(s\,i)) = f(s\,i)$ (the map $1 \to s$ picking i). By the universal property of $T(s)$ it is $\Phi_{f\circ s}$. □

6. The unit is injective

Theorem 6.1. *If X is a set, $\eta : X \to F(X)$ is injective.*

Proof. Let $\eta a = \eta b$. By effectiveness there merely is a link: a stage t and m, n from the one-point stages a and b with $m_*(\mathrm{gen}) = n_*(\mathrm{gen})$, i.e. the generators $m(*)$ and $n(*)$ of $T(t)$ are equal. Generators are distinct, so $m(*) = n(*)$, and $a = t(m(*)) = t(n(*)) = b$. Since X is a set, $a = b$ is a proposition and the link may be used although it exists only merely. □

7. Size

Let E be a set in a universe U' which is *locally small* with respect to a smaller universe U: each identity type $a =_E b$ is equivalent to a type in U. A type is *small* if it is equivalent to a type in U. The finite stages, and the groups $T(c)$, are in U.

Lemma 7.1. *For elements $x = (s,u)$ and $y = (s',v)$, with $s + s' = (c,g)$,*

$$[x] = [y] \iff \left\| \textstyle\sum_{r:\mathrm{El}(c)\to\mathrm{El}(c)} \left(\prod_z g(rz) = g(z)\right) \times (r\,\mathrm{inl})_* u = (r\,\mathrm{inr})_* v \right\|.$$

The right side is small.

Proof. From right to left, $(s + s', r\,\mathrm{inl}, r\,\mathrm{inr})$ is a link. From left to right, take a link (t, m, n) and let $\rho = [m,n] : \mathrm{El}(c) \to t$. If $\mathrm{El}(c)$ is empty, $T(c)$ is trivial and $r = \mathrm{id}$ works. Otherwise choose a point d and, for $z \in t$, let $\sigma(z)$ be some i with $\rho(i) = z$ if there is one (decided on the finite type $\mathrm{El}(c)$), and d otherwise. Then $r = \sigma \circ \rho$ satisfies $\rho \circ r = \rho$, hence $g \circ r = g$, and $(r\,\mathrm{inl})_* u = \sigma_* m_* u = \sigma_* n_* v = (r\,\mathrm{inr})_* v$. The right side involves a function between finite types, equalities in E, which are small, and an equality in $T(c)$, which is in U. □

Lemma 7.2. *Being small is a proposition.*

Proof. Given (Y, e) and (Y', e') with $e : Y \simeq B$, $e' : Y' \simeq B$, the equivalence $d = e'^{-1} \circ e : Y \simeq Y'$ satisfies $e' \circ d = e$. By univalence for U (induction on the equivalence d) it suffices to treat $Y' = Y$ and $d = \mathrm{id}$, where $e' = e$ because equivalences with the same underlying map are equal (being an equivalence is a proposition). □

Theorem 7.3. *If E is a locally small set, then $F(E)$ is locally small and the fibres of η are small: for $z = [(s,u)]$,*

$$\left(\textstyle\sum_{a:E} \eta a = z\right) \;\simeq\; \left\| \textstyle\sum_{i:\mathrm{El}(s)} \eta(s\,i) = z \right\|.$$

The same holds for $A(E)$.

Proof. On representatives, local smallness is Lemma 7.1. For the fibre: if $\eta a = z$, a link from the one-point stage a to (s,u) gives a stage t and maps m, n with $\mathrm{gen}_{m(*)} = n_* u$. By the support property $m(*) = n(i)$ for some $i \in \mathrm{El}(s)$, and $a = t(m(*)) = t(n(i)) = s(i)$. The fibre is a proposition by Theorem 6.1, and the right side is small by Lemma 7.1. Every element of $F(E)$ is merely a representative, and by Lemma 7.2 the statements are propositions, so they hold at every element. □

8. The free abelian group

Take $T(c) = \mathbb{Z}^c$ in Section 5 and let $A(X)$ be the resulting colimit. An element of $A(X)$ is represented by a stage $f : \mathrm{El}(c) \to X$ and an integer vector $u \in \mathbb{Z}^c$, read as the formal sum $\sum_i u(i)\, f(i)$; the unit is $\eta(x) = [(1, x), 1]$.

Theorem 8.1. *$A(X)$ is an abelian group, and it is the free abelian group on X: every map $X \to G$ to an abelian group extends uniquely to a homomorphism $A(X) \to G$. If X is a set, η is injective; if X is a locally small set, $A(X)$ is locally small and η has small fibres.*

Proof. Commutativity is in Theorem 5.3 (the swap of stages); then $A(X)$ is a target, and Theorems 5.4, 6.1 and 7.3 apply with Propositions 4.3 and 4.4. □

Unfolding Lemma 7.1 for this instance describes equality in $A(E)$ by integer vectors. An endomap r of $\mathrm{El}(c + c')$ with $[f, f'] \circ r = [f, f']$ identifies formal variables which have the same value in E; the condition $(r \,\mathrm{inl})_* u = (r \,\mathrm{inr})_* v$ says that for every w,

$$\sum_{r(\mathrm{inl}\, i) = w} u(i) = \sum_{r(\mathrm{inr}\, j) = w} v(j).$$

Corollary 8.2. *$\sum_i u(i)\, f(i) = \sum_j v(j)\, f'(j)$ in $A(E)$ if and only if there merely is a partition of the formal variables of the two sides into classes on which $[f, f']$ is constant, such that in each class the coefficients of the left side and of the right side have the same total.*

Proof. A map r as above gives the partition into its fibres, with the class of w having the totals of the fibre over $r(w)$. Conversely, a partition (a decidable equivalence relation) gives r, sending each variable to the first element of its class in the order of the code; this choice depends only on the class, so the fibres of r are the classes. □

Corollary 8.3. *Equality $\sum_i u(i)\, f(i) = \sum_j v(j)\, f'(j)$ in $A(E)$ is a finite disjunction of conjunctions of equalities in E: it holds iff merely, for one of the finitely many maps $r : \mathrm{El}(c+c') \to \mathrm{El}(c + c')$ whose fibre sums agree (a decidable condition), $[f, f'](r(z)) = [f, f'](z)$ for every z.*

Proof. The maps $\mathrm{El}(a) \to \mathrm{El}(b)$ are coded by a finite type $\mathrm{El}(\mathrm{fn}(a, b))$, with $\mathrm{fn}(0, b) = 1$, $\mathrm{fn}(1, b) = b$ and $\mathrm{fn}(a + a', b) = \mathrm{fn}(a, b) \times \mathrm{fn}(a', b)$, by a decoding which is surjective up to pointwise equality; the condition on the fibre sums is a finite conjunction of equalities of integers. □

Examples. Let $e_0, e_1, e_2 \in E$.

(1) $e_0 - e_1 + 2e_2 = e_0 + e_1$ holds iff $e_1 = e_2$. Write the left side as $X_0 - X_1 + 2X_2$ and the right side as $Y_0 + Y_1$, with $X_0, Y_0 \mapsto e_0$ and $X_1, Y_1 \mapsto e_1$, $X_2 \mapsto e_2$. Counting the right side negatively, the coefficients are $1, -1, 2, -1, -1$, and every class must have total 0. The only negative coefficients are the three -1's, so the class of X_2 contains two of X_1, Y_0, Y_1 (or all other variables); in either case it contains X_1 or Y_1, whose value is e_1, so $e_1 = e_2$. Conversely, if $e_1 = e_2$ the classes $\{X_0, Y_0\}$ and $\{X_1, X_2, Y_1\}$ work.

(2) The fibre of η at $e_0 - e_1 + e_2$ is $\|e_0 = e_1 \vee e_1 = e_2\|$. By Theorem 7.3 an element in the fibre is one of e_0, e_1, e_2, and $\eta(e_i) = e_0 - e_1 + e_2$ asks for a partition of Y, X_0, X_1, X_2 with coefficients $-1, 1, -1, 1$ and all totals 0, with $Y \mapsto e_i$. For $i = 0$ the possibilities need $e_1 = e_2$ (classes $\{Y, X_0\}, \{X_1, X_2\}$) or $e_0 = e_1 = e_2$; for $i = 2$, $e_0 = e_1$ or $e_0 = e_1 = e_2$; for $i = 1$, $e_0 = e_1 = e_2$.

(3) The fibre of η at $e_0 - e_1 + 2e_2$ is empty. Here no partition is needed: the augmentation $A(E) \to \mathbb{Z}$, the homomorphism sending every generator to 1, sends $e_0 - e_1 + 2e_2$ to 2 and every $\eta(a)$ to 1.

9. What is formalised

Everything above is checked in Unit, in `unit/sag/freegroup`. The development has the parameters of `sag/`: univalence and replacement (images and their surjectivity). Function extensionality, propositional truncation and effective set quotients are derived from them. There are no new axioms and no higher inductive types.

Section	Files
2	`Decisions`, `Codes`
3	`Words`, `Functor`, `Universal`
4	`FinSum`, `Multiples`, `AbelianInstance`
5	`Interface`, `Stages`, `Colimit`, `FreeGroup`, `Lift`, `FreeInstance`
6	`Injective`
7	`Identification`, `SmallProp`, `Small`
8	`AbelianInstance`, `Abelian`, `AbelianEquality`, `AbelianExamples`

The list of properties at the beginning of Section 5 is the record `FinFree`; the colimit files are written for an arbitrary `FinFree`, and `FreeInstance.free` and `AbelianInstance.abelian` are the two instances. The final statements are in `Theorems.seg`, with explicit types: `free-group`, `free-lift-mul`, `free-lift-eta`, `free-lift-unique`, `free-eta-inj`, `free-locally-small`, `free-small-fibres`, and the same with prefix `ab-` for abelian groups, together with `ab-comm`. The universal property is stated as existence of the homomorphism and pointwise uniqueness. For $A(X)$, `ab-eq-identification`, `ab-eq-partition` and `ab-eq-finite` are the description by maps r with equal fibre sums, Corollary 8.2 and Corollary 8.3; `ab-example1`, `ab-example2`, `ab-example3` are the three examples, for a set E. In the formal proofs of the first two examples the partitions are not enumerated: from a map r, the fibre sums at one or two points are integer expressions in the decidable memberships $[r(Z) = w]$, and a case analysis on these memberships shows which variables share a class. The third is the augmentation, an instance of the universal property with $G = \mathbb{Z}$.

Two universe levels are used: the finite stages live in U_l and the generators in U_k with $l \leq k$, so that for E in U_{l+1} locally small the groups $T(c)$ stay in U_l. The integers, with the arithmetic of $\mathbb{N}$ they use, are those of `sag/stoneseq`; that being an equivalence is a proposition is `sag/em/EquivProp`.

The development consists of 24 new files (2777 lines) and links 13 existing files. Checking it takes 0.22 s (1255 declarations); no declaration unfolds more than 42 definitions (the most are the explicit fibre sums of the examples).

References

[1] B. L. van der Waerden, *Free products of groups*, American Journal of Mathematics **70** (1948), 527–528.